

Nam-Jin Park

Assistant Professor

Department of Control & Instrumentation Engineering,

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Research Interests

Principle Topics

- Control theory
- Multi-agent system
- Robotic & Automation system

Application 1: Multi-Agent Control Systems

- Distance and position-based formation control for multi-agent systems
- Onboard localization for multi-agent systems

Application 2: Robotic Automation Systems

- Multi-drone system for cooperative localizations
- Delta-robot-based intelligent pick & place systems

Education

Gwangju Institute Science and Technology — *Gwangju, Korea*

M.S./Ph.D., Mechanical Engineering, Top Student Award — *Aug. 2024*

- **Thesis:** Controllability and Controllable Subspaces of Structured Networks: A Graph-Theoretical Approach
- **Advisor:** Prof. Hyo-Sung Ahn

Pukyong National University — *Busan, Korea*

B.S., Control and Instrumentation Engineering — *Feb. 2018*

Professional Experience

Pukyong National University — *Busan, Korea*

Assistant Professor - Department of Control and Instrumentation Engineering — *Mar. 2026 - Current*

Kunsan National University — *Jeonbuk, Korea*

Assistant Professor - Department of Electrical Engineering — *Mar. 2025 - Feb. 2026*

Gwangju Institute Science and Technology — *Gwangju, Korea*

Postdoctoral Fellow - Distributed Control and Autonomous Systems Lab. — *Aug. 2024 - Feb. 2025*

Publications

Journal Under Review (International)

* **Corresponding Author**

8. **N.-J. Park**, L. C. G. Lebon, W.-Y. Choi, and Q. V. Tran*, "Structural and Strong Structural Controllability of Matrix-Weighted Networks".
7. J.-M. Ma, K.-K. Oh, H.-S. Ahn, and **N.-J. Park***, "Targeted Topological Clustering in Directed Consensus Networks: Minimum-Cost Design of Edge Modifications".
6. J.-H. Yang, S.-W. Kang, and **N.-J. Park***, "An Online Matheuristic Scheduling Framework for Multi-Robot Pick-and-Place Systems".
5. **N.-J. Park**, W.-Y. Choi, and K.-H. Lee*, "Target Controllability of Hierarchical Directed Acyclic Networks: A Walk Family Characterization and Polynomial-Time Menger Dual".
4. S.-B. Won, Y.-U. Kim, and **N.-J. Park***, "A Construction of Connected Dominating Set with Predetermined Nodes", *International Journal of Control, Automation, and Systems* **- Accepted (2026.09.09)**.
3. **N.-J. Park**, Q. V. Tran, and Y.-B. Bae*, "Cluster Target Controllability in Multi-Agent Systems: Bridging Consensus and Functional Target Control".

2. S.-B. Won, Y.-U. Kim, H.-S. Ahn, and **N.-J. Park***, "FoRCE: Scalable Energy-Efficient Target Control of Complex Networks".

1. Y.-U. Kim, S.-B. Won, H.-S. Ahn, and **N.-J. Park***, "Energy-Optimal Target Control via Edge Optimization on Structured Networks".

Journal Under Review (Domestic)

2. **N.-J. Park** and K.-H. Oh*, "Structural Identifiability of Leader-Cluster Consensus Values from Follower-Cluster Measurements".

1. **N.-J. Park** and K.-H. Oh*, "Cluster-based Controllability in Structured Networks", *Journal of Institute of Control, Robotics and Systems* **- Accepted (2026.09)**.

Journal Publications (International)

14. T.-W. Um, S.-W. Kang, K.-H. Kim, G.-H. Kim, H.-S. Ahn, W.-Y. Choi, and **N.-J. Park***, "Deployable Onboard Vision-Based Pick-and-Place Automation System for Unstructured Environments", *IEEE Robotics and Automation Letters*, 2026.

13. K.-H. Lee and **N.-J. Park***, "An exact gamma–beta scale-mixture law for the squared norm of a complex-Gaussian-scaled sum of hyperspherical random vectors", *AIMS Mathematics*, 2026 **- JCR: 8.8%**.

12. J.-H. Son, J.-M. Ma, H.-G. Lee, **N.-J. Park**, and H.-S. Ahn*, "Ad-hoc Network for Unmanned Mobile Vehicles: Implementation and Experiments", *IEEE Access*, 2026.

11. **N.-J. Park**, Y.-U. Kim, K.-H. Oh and H.-S. Ahn*, "Controllable Subspaces in Structured Networks of Hierarchical Directed Acyclic Graphs: Controllability of Individual Nodes", *IEEE Transactions on Automatic Control*, 2026 **- JCR: 8.8% (IF 7.0)**.

10. H.-G. Lee, J.-M. Ma, **N.-J. Park**, and H.-S. Ahn*, "A Distributed Influence Measurement Algorithm in Leader-Follower Networks", *Systems & Control Letters*, 2025 **- JCR: 42%**.

9. **N.-J. Park**, Y.-U. Kim, and H.-S. Ahn*, "Bio-inspired projected gradient method for leader selection in minimum cost problem", *Artificial Life and Robotics*, 2025 **- SCOPUS (Invited Paper)**.

8. **N.-J. Park**, Y.-U. Kim, and H.-S. Ahn*, "A Distributed Framework for the Minimum Cost Problem Based on Graph Decomposition", *Journal of the Franklin Institute*, 2025 **- JCR: 8.5%**.

7. **N.-J. Park**, S.-H. Kwon, Y.-B. Bae, B.-Y. Kim, Kevin L. Moore, and H.-S. Ahn*, "Merging Rules for Strong Structural Controllability and Minimum Input Problem in Undirected Networks", *Journal of the Franklin Institute, 2025* **- JCR: 8.5%**.

6. **N.-J. Park** and H.-S. Ahn*, "Network Controllability of Structured Networks: A Survey on Graph-Theoretical Approaches", *SICE Journal of Control, Measurement, and System Integration*, 2025 **- SCOPUS (Invited Paper)**.

5. **N.-J. Park**, K.-H. Kim, J.-M. Ma, J.-H. Son, H.-S. Ahn*, "Implementation of a Multi-Drone System for UWB-Based Onboard Localization with Free Structural Anchors", *Mechatronics*, 2024 **- JCR: 29.8%**.

4. **N.-J. Park**, Y.-U. Kim, and H.-S. Ahn*, "Fixed node determination and analysis in directed acyclic graphs of structured networks", *Journal of the Franklin Institute*, 2024 **- JCR: 8.5%**.

3. **N.-J. Park**, Y.-U. Kim, H.-S. Ahn*, and Kevin L. Moore, "Strong Structural Controllability and Observability of Individual Nodes: A Graph-Theoretical Approach", *IEEE Transactions on Automatic Control*, 2024 **- JCR: 8.8% (IF 7.0)**.

2. J.-M. Ma, **N.-J. Park**, H.-G. Lee, and H.-S. Ahn*, "Comments on On multi-consensus and almost equitable graph partitions [Automatica 103 (2019) 53–61]", *Automatica*, 2024 **- JCR: 14.6%**.

1. K.-H. Kim, **N.-J. Park**, H.-G. Lee, and H.-S. Ahn*, "3-D Localization With Coplanar Anchors", *IEEE Communications Letters*, 2023 **- JCR: 32.1%**.

Conference Proceedings (International)

4. S.-W. Kang, **N.-J. Park**, T.-W. Um, H.-S. Ahn*, "Implementation of Intelligent Robotic Systems for Pick and Place based on Image Processing", *14th Asian Control Conference* (*ASCC*), 2024.

3. **N.-J. Park**, Y.-B. Bae, Kevin L. Moore, and H.-S. Ahn*, "Controllability Matrix Analysis of Structured Networks: A Tight Lower Bound on the Dimension of Controllable Subspace", *22nd European Control Conference* (*ECC*), 2024.

2. **N.-J. Park**, Y.-U. Kim, and H.-S. Ahn*, "Strong Structural Controllability of Directed Graphs via Zero Forcing Sets", *22nd IFAC World Congress*, 2023.

1. Y.-U. Kim, **N.-J. Park**, J.-W. Lee, and H.-S. Ahn*, "Distributed Continuous-Time Newton Method via Blended Dynamics", *9th IFAC Conference on Networked Systems NECSYS*, 2022.

Conference Proceedings (Domestic)

2. **N.-J. Park**, H.-G. Lee, and H.-S. Ahn*, "Building Temperature Control using Control Centrality", in *20th International Conference on Control, Automation and Systems* (*ICCAS*), 2020.

1. S.-H. Park, **N.-J. Park**, and H.-S. Ahn*, "Distance-based Localization using Ultra-Wideband Sensor and Implementation of a UAV Control Platform", in *34th Institute of Control, Robotics, and Systems* (*ICROS*) *Conference*, 2019.

Research Experiences

Growth Pathway Support Program for Outstanding Talent in Science and Engineering

Busan, Korea

Funded by the Ministry of Education through Pukyong National University (₩80,000,000)

Sep. 2026 - Present

- **Role:** Principal Investigator

Development of a Self-Protective Smart Battery Module — *Jeonbuk, Korea*

Funded by the Kunsan National University (₩50,000,000) — *Oct. 2025 - Feb. 2026*

- **Role:** Principal Investigator

Development of AI-based Dynamic Graph Generation and UAM Autonomous Operation Optimization System with Real-time Data — *Gwangju, Korea*

Funded by the Gwangju Institute Science and Technology (₩30,000,000) — *Jan. 2025 - Feb. 2025*

- **Role:** Principal Investigator

Establishment and Core Technology Development for Fresh Produce Fulfillment Center — *Gwangju, Korea*

Funded by the Korea Institute of Planning and Evaluation for Technology in Food, Agriculture, and Forestry (₩800,000,000) — *Apr. 2022 - Feb. 2025*

- **Role:** Principal Manager

Development of Autonomous Operation Technology for Multi-UAVs for Stable Control of End-Effector Drones — *Gwangju, Korea*

Funded by the National Research Foundation of Korea (₩123,030,000) — *Mar. 2022 - Feb. 2025*

- **Role:** Researcher

Development of Autonomous Formation and Collaborative Control for Multi-UAVs — *Gwangju, Korea*

Funded by the National Research Foundation of Korea (₩125,000,000) — *Sep. 2018 - Feb. 2022*

- **Role:** Principal Manager

Formation Control of Multi-Agent Systems with Robustness and Security — *Gwangju, Korea*

Funded by the National Research Foundation of Korea (₩100,000,000) — *Oct. 2019 - Feb. 2021*

- **Role:** Researcher

Development of Next-Generation 3D Perception Composite Sensor Module Integrating 3D Image Sensors and Micro LiDAR — *Gwangju, Korea*

Funded by the Korea Aerospace Research Institute (₩170,000,000) — *Sep. 2018 - Jul. 2019*

- **Role:** Researcher

Patents

Application (Domestic) — * Primary Inventor

1. **N.-J. Park***, "자동복구 기능을 갖는 보호회로 및 이를 포함하는 배터리 시스템", KR pending, 10-2026-0030922, 2026.

Application (International) — * Primary Inventor

1. **N.-J. Park***, K.-H. Kim, J.-M. Ma, J.-O. Park, J.-H. Son, H.-G. Lee, and H.-S. Ahn, "Unmanned Aerial Vehicle Formation Control System and Method Therefor", US, 18/293,420, 2024 **- Allowed (2026.10)**.

Registration (Domestic) — * Primary Inventor

4. H.-G. Lee, **N.-J. Park**, S.-H. Kwon, Y.-H. John, and H.-S. Ahn, "무인 비행체의 측위 성능 분석 시스템 및 방법", KR, 10-2847977, 2025.
3. **N.-J. Park***, K.-H. Kim, J.-M. Ma, H.-G. Lee, J.-G. Lee, and H.-S. Ahn, "다중 무인 비행시스템 및 그 시스템의 제어방법", KR, 10-2630845, 2024.
2. **N.-J. Park***, K.-H. Kim, J.-M. Ma, J.-O. Park, J.-H. Son, H.-G. Lee, and H.-S. Ahn, "무인 비행체 편대 제어 시스템 및 그 방법", KR, 10-2573312, 2023.
1. K.-H. Kim, **N.-J. Park**, J.-M. Ma, H.-G. Lee, J.-O. Park, J.-H. Son, and H.-S. Ahn, "위치 측정 방법", KR, 10-2591036, 2023.

Professional Services

Journal Editors

International Journal of Control, Automation, and Systems (IJCAS)

Associate Editor — *Dec. 2025 - Present*

Journal Reviewers

- IEEE Transactions on Automatic Control (ISSN: 0018-9286)
- IEEE Transactions on Control of Network Systems (ISSN: 2325-5870)
- IEEE Transactions on Network Science and Engineering (ISSN: 2327-4697)
- Automatica (ISSN: 0005-1098)
- Systems & Control Letters (ISSN: 0167-6911)
- Journal of the Franklin Institute (ISSN: 0016-0032)
- IET Control Theory & Applications (ISSN: 1751-8644)
- IEEE Systems Journal (ISSN: 1932-8184)
- International Journal of Control, Automation, and Systems (ISSN: 1598-6446)
- Journal of Supercomputing (ISSN: 0920-8542)

Conference Reviewers

- Conference on Decision and Control
- Asian Control Conference
- European Control Conference
- Chinese Control and Decision Conference
- IFAC World Congress

Teaching Experience

Gwangju Institute Science and Technology

Gwangju, Korea

Teaching Assistant

Mar. 2021 - Dec. 2023

- General Physics Lecture
- General Physics Experiment

Kunsan National University

Jeonbuk, Korea

Assistant Professor

Mar. 2025 - Feb. 2026

- Control Engineering
- Modern Control Engineering
- Logical Circuit Design
- Microprocessor
- Artificial Intelligence

Pukyong National University

Busan, Korea

Assistant Professor

Mar. 2026 - Present

- Hardware System Design
- Electromagnetics
- Linear Algebra
- Linear Systems
- Applied Robotics
- Capstone Design